Session 18

Dart Classes & Objects Assignment

Objective: Practice creating classes, constructors, methods, optional parameters, and object-oriented programming concepts in Dart.

Task 1: Restaurant Class

Description: Create a Restaurant class with fields **name**, **cuisine**, and **rating**.

```
class Restaurant {
  String name;
  String cuisine;
  double rating;

  Restaurant(this.name, this.cuisine, this.rating);
}

void main() {
  Restaurant r =
      Restaurant("Hocco","Fast Food",4.6);

  print("Name: ${r.name}");
  print("Cuisine: ${r.cuisine}");
  print("Rating: ${r.rating}");
}
```

Expected Output:
Name: Hocco
Cuisine: Fast Food
Rating: 4.6

Task 2: Movie Class

Description: Create a Movie class and instantiate two movie objects.

```
class Movie {
  String title;
  String genre;
  int releaseYear;

  Movie(this.title,this.genre,this.releaseYear);
}

void main() {
  Movie m1=Movie("3 Idiots","Comedy",2009);
  Movie m2=Movie("Interstellar","Sci-Fi",2014);

  print("${m1.title} | ${m1.genre} | ${m1.releaseYear}");
  print("${m2.title} | ${m2.genre} | ${m2.releaseYear}");
}
```

Expected Output:
3 Idiots | Comedy | 2009
Interstellar | Sci-Fi | 2014

Task 3: Product Class

Description: Create a Product class with a displayProduct() method.

```
class Product {
  String productName;
  double price;
  bool isAvailable;

  Product(this.productName,this.price,this.isAvailable);

  void displayProduct() {
    print("Product: $productName");
    print("Price: ■$price");
    print("Available: $isAvailable");
  }
}

void main() {
  Product p=Product("Wireless Earbuds",1499,true);
  p.displayProduct();
}
```

Expected Output:
Product: Wireless Earbuds
Price: ■1499.0
Available: true

Task 4: Optional Parameters

Description: Refactor the Movie class using named parameters, nullable genre, and a default release year.

```
class Movie {
  String title;
  String? genre;
  int releaseYear;

  Movie({
    required this.title,
    this.genre,
    this.releaseYear = 2024,
  });
}

void main() {
  Movie movie = Movie(title: "Kalki");
  print("${movie.title} | ${movie.genre} | ${movie.releaseYear}");
}
```

Expected Output:
Kalki | null | 2024

Task 5: Playlist Class

Description: Create a Playlist class with an addSong() method that increments the song count.

```dart
class Playlist {
  String playlistName;
  int songCount;

  Playlist(this.playlistName,this.songCount);

  void addSong() {
    songCount++;
  }
}

void main() {
  Playlist p=Playlist("Workout Mix",0);

  p.addSong();
  p.addSong();
  p.addSong();

  print("Songs: ${p.songCount}");
}
```

Expected Output:
Songs: 3

Conclusion: This session covered Dart classes, objects, constructors, methods, optional named parameters, default values, and basic object-oriented programming concepts.